Solar Package: reusable templates and optional battery

A system is a Package record. Staff can add future systems and component groups without changing the calculation program.

Solar Package — parent fields

Field / internal field name	Frappe type / linked DocType	Who fills it?	Example / purpose
Package Name package_title	Data	Staff enters	A named system profile, including any compatible hybrid combination.
Brand brand	Link → Brand	Staff selects existing metadata	Brand used for display / existing print content.
System Type system_type	Select	Staff selects existing metadata	Micro / String / Hybrid.
Grid Type grid_type	Select	Staff selects existing metadata	On-Grid / Off-Grid / Hybrid.
Inverter Count Formula inverter_count_formula	Small Text — expression	Staff enters per package	panel_count; ceil(panel_count / 2); or a count rule appropriate to this system.
Allow Battery allow_battery	Check	Staff enables only for compatible hybrid packages	When false, the proposal has include_battery = false.
Component Templates component_rules	Table → Solar Package Component Rule	Staff selects templates and optional conditions	Downrun, inverter links, looping, common protection, DB templates and optional battery templates.
Additional Input Definitions input_definitions	Table → Solar Package Input	Staff adds inputs and defaults	Lengths, installation choices and any other variables used by the rules.
Disabled disabled	Check	Optional staff choice	Stops selection for new proposals; does not change saved proposals.

Solar Package Component Rule — child table of Package

Field / internal field name	Frappe type / linked DocType	Who fills it?	Example / purpose
Template Alias template_alias	Data	Staff enters a stable unique key within the package	downrun / inverter_links / common / battery_db.
Component Template component_template	Link → Solar Component Template	Staff selects a reusable saved template	A Downrun or Common Protection template.
Include When include_when	Small Text — boolean expression	Blank = always; staff may enter a condition	include_battery enables the whole battery template.
Enabled enabled	Check	Defaults Yes; staff can turn off a mapping	Disabled mappings contribute zero quantities.

Phase and cabling conditions can be entered here for an entire template, or on individual material rows. Multiple templates can contribute to the same display group. Do not require a fixed number of groups.

Example: package template mappings — names are illustrative

Alias	Template selected	Include When	Meaning
downrun	A suitable Downrun template	Always	Uses the phase/cabling inputs where its rows need them.
inverter_links	Micro Inverter to Downrun	Always	Inverter cables / accessories for this package.
db_loop	Downrun to DB Looping	Always	Phase-specific looping cables.
common	Common Protection	Always	Reusable common materials.
battery	Battery Materials	include_battery	Only selected when this compatible hybrid includes a battery.
battery_db	Battery DB	include_battery	Uses the same optional battery choice.

Battery is an add-on: staff enable Allow Battery on the applicable hybrid packages, then add conditional Battery / Battery DB template links. Changing to an incompatible package clears the battery choice before generation.

Aliases are stable reference names. For example, qty("downrun.clamp") means the clamp row in the template attached with alias downrun. Reusing a template in another package does not depend on its display name or visual row order.

No template header or package requires a manufactured Item or manufacturing BOM. Existing legacy records keep their saved data and calculation behavior.

Package-defined inputs: defaults staff can extend

The five main selections remain. Each package can define extra lengths and installation choices; their values are saved on the proposal.

Solar Package Input — child table of Package

Field / internal field name	Frappe type / linked DocType	Who fills it?	Example / purpose
Variable Key variable_key	Data	Staff enters once	db_loop_length; unique and different from core variable / helper names.
Label label	Data	Staff enters	DB Looping Length.
Input Type input_type	Select	Staff selects	Int / Float / Check / Select / Data.
Unit uom	Link → UOM	Optional staff selection	Meter for a route length.
Default Value default_value	Data	Staff enters a literal of the chosen type	8 for Float, or 0 / 1 for Check. A default is not a formula.
Choices select_options	Small Text	For Select inputs only	One permitted choice per line.
Show When show_when	Small Text — boolean expression	Blank = always; optional condition	include_battery for a battery quantity input.
Required required	Check	Staff chooses	A visible required input must have a valid value before its rule is evaluated.

Example definitions entered by staff

Variable key	Type / unit	Default	Use / visibility
downrun_length	Float / Meter	A default appropriate to the package	The route length used by that package; staff may change it for a site.
db_loop_length	Float / Meter	8 in this example	Quantity of each selected DB looping conductor.
db_meter_distance	Float / Meter	Blank; required when the selected rules need it	The above-3-metre condition in Package.md.
self_screws	Check	Staff-defined 0 or 1	Whether wall plugs are needed.
battery_count	Int	1 in this example	Shown only when include_battery is true.

Solar Proposal Input Value — child table of Proposal

Field / internal field name	Frappe type / linked DocType	Who fills it?	Example / purpose
Variable Key variable_key	Data — saved copy	Auto from package definition	db_loop_length.
Label / Type / Unit label / input_type / uom	Data / Select / Link → UOM	Auto; saved with the proposal	DB Looping Length / Float / Meter.
Choices / Show When / Required select_options / show_when / required	Small Text / Small Text / Check	Saved copies of the input definition	Existing proposals keep their original choices, visibility and validation when a package definition changes.
Value value	Data	Defaults once; staff edits per proposal	8. Server converts and validates it as Float from the saved input_type.

The stored value field has one Frappe fieldtype. The form can display a number, checkbox or choices, while the server parses the value according to its input definition. Template/default edits do not rewrite saved proposal inputs.

Core variables available to conditions and formulas

Expression variable	Comes from	Example	Staff can redefine it?
phase	Proposal.connection_type	Three Phase	No; select it on the proposal.
cabling_type	Proposal.down_run_cabling	Armoured Cabling / Conduit Cabling	No; select it on the proposal.
capacity_kw / panel_capacity_w	The two capacity inputs	3 kW / 660 W	No; enter values on the proposal.
panel_count	ceil(capacity_kw × 1000 / panel_capacity_w)	5	No; calculated by the server.
inverter_count	The selected package inverter-count formula	5 or 3, depending on package	Edit the package formula, not this saved result.
include_battery	Proposal battery choice and package compatibility	True / False	No; choose it on a compatible proposal.

Inverter-count formulas may reference core inputs, panel_count and package-defined inputs. They may not reference component quantities, which are calculated after inverter_count. Missing required values are errors, not assumed zeroes. Regeneration preserves existing site-entered values, refreshes their definition metadata and applies defaults only to newly introduced keys; incompatible values are reported for staff to correct.

Component templates: attach the Item and the rule to the same row

Generic names describe a component. Staff choose its actual ERPNext Item once here, and proposals use that Item automatically.

Solar Component Template — parent fields

Field / internal field name	Frappe type / linked DocType	Who fills it?	Example / purpose
Template Name template_name	Data	Staff enters	A reusable name such as Common Protection or a particular Downrun variant.
Display Group group_name	Data	Staff enters an editable group name	Downrun / Micro Inverter to Downrun / Downrun to DB Looping / Common for all / another group.
Material Rules items	Table → Solar Component Template Item	Staff adds / removes / edits rows anytime	Each row selects one actual Item and contains its inclusion and quantity rules.
Validation Condition validation_condition	Small Text — optional boolean expression	Optional staff-defined supported-input check	Validate a configured range instead of extrapolating beyond the defined rules.
Validation Message validation_message	Small Text	Staff explains a failed validation condition	A clear instruction when this template does not support the selected inputs.
Notes / Disabled notes / disabled	Small Text / Check	Optional staff entries	Notes and optional retirement; no approval workflow. Empty templates are allowed.

Solar Component Template Item — material-rule child fields

Field / internal field name	Frappe type / linked DocType	Who fills it?	Example / purpose
Rule Key rule_key	Data	Staff enters a stable unique key	yellow_ac_cable; used by references, not the editable label or row index.
Generic Component Label component_label	Data	Staff enters	Yellow AC cable, 6 sq.mm.
Actual ERPNext Item item_code	Link → Item	Staff selects once in the template	The exact Item to supply and value. Do not guess from the generic name.
UOM uom	Link → UOM	Defaults from Item; staff confirms	The quantity formula returns this unit, for example Meter.
Include When include_when	Small Text — boolean expression	Blank = always; staff may edit	phase == "Three Phase".
Quantity Formula quantity_formula	Small Text — numeric expression	Staff enters a number or equation	8; db_loop_length; ceil(inverter_count / 7); qty("clamp") * 2.
Specification description	Small Text	Defaults from Item; optional template-specific text	Saved in the generated material row for printing.
Enabled enabled	Check	Defaults Yes; staff may disable	A disabled row generates zero quantity.
Notes notes	Small Text	Optional staff explanation	Why the item is included and what its formula means.

Example: replace the selected Item with another suitable actual Item while keeping component_label, include_when and quantity_formula. The rule stays attached to the material role; UOM and conversion must still be valid.

Staff actions provided by the template editor

Action	Staff does	System shows	Stored result
Choose Item	Select an actual Item from ERPNext	Item name, stock UOM and description	A direct Item link, not a name-matching search rule.
Build a condition	Pick a variable, comparison and value; combine conditions if needed	The equivalent readable boolean expression	include_when on the material row or package-template link.
Enter a quantity	Enter a constant, arithmetic or approved helper	Available variables and existing quantity keys	quantity_formula on the material row.
Test Rules	Supply sample inputs	Included / excluded rows, substitutions, calculated quantity and errors	No proposal changes until successful generation is applied.

No separate generic Component master is required for this stage. Reusable templates share their Item choices across packages. Each enabled material rule must select an actual Item before it can generate a priced proposal; staff may keep an empty template and add its rules later.

How staff enter conditions and dependent quantity equations

Examples show the structure of editable data. Actual Items and electrical specifications are chosen by staff, not seeded by the program.

Example: Downrun to DB Looping template

Rule key / generic label	Actual Item / UOM	Include When	Quantity Formula
red_ac_cable / Red AC cable, 6 sq.mm	Staff-selected red Item / Meter	Always	db_loop_length
black_ac_cable / Black AC cable, 6 sq.mm	Staff-selected black Item / Meter	Always	db_loop_length
yellow_ac_cable / Yellow AC cable, 6 sq.mm	Staff-selected yellow Item / Meter	phase == "Three Phase"	db_loop_length
blue_ac_cable / Blue AC cable, 6 sq.mm	Staff-selected blue Item / Meter	phase == "Three Phase"	db_loop_length

With db_loop_length = 8, Single Phase generates red and black at 8 metres each. Three Phase generates all four at 8 metres each. The form supplies phase; the additional-input table supplies db_loop_length.

Example: dependent quantities within a Downrun template

Rule key	Quantity Formula	Required information	Result for the example
ug_cable	downrun_length	downrun_length = 25 metres	25 metres
clamp	qty("ug_cable") * 2	Calculated cable quantity	50 clamps
screw	qty("clamp") * 2	Calculated clamp quantity	100 screws
wall_plug	0 if self_screws else qty("screw")	Screw quantity and self_screws input	0 for self screws, otherwise 100
q_terminal (in an inverter template)	ceil(inverter_count / 7)	Calculated inverter count	7 inverters → 1; 8 inverters → 2

qty("clamp") refers to a rule key within the same template instance. qty("downrun.clamp") uses the package mapping alias to refer to a row in another template. Conditions and formulas both participate in dependency checking.

Supported expressions and required validation

Feature	Example / behavior	What staff can change	What the program checks
Arithmetic and rounding	ceil(downrun_length / 3), min(...), max(...)	Numeric constants, variables and approved arithmetic	Numeric, finite, non-negative quantities and whole-number UOM rules.
Conditions	phase == "Three Phase" and db_meter_distance > 3	Comparisons, and / or / not, and conditional quantities	Boolean result, known variables and valid required inputs.
Supported ranges	Template validation plus non-overlapping range conditions	The input ranges and formulas staff have defined	Report an unsupported range; do not silently invent an equation outside it.
Dependencies	Cable → clamp → screw → wall plug	References using stable aliases / rule keys	Unknown references and cycles are errors. Excluded known rows/templates resolve to zero.
Expression limits	Only approved variables and helpers	Formula data; no server scripting	No imports, attributes, database access or unrestricted Python evaluation.

Route length is separate from total conductor quantity. Red 35 m plus black 35 m does not automatically make a 70 m conduit route. Staff should reference the intended route input explicitly.

Acceptance examples for the first implementation

Scenario	Expected result
Staff changes Item selection or a template equation	The next generation uses it; saved proposals remain unchanged.
Staff adds a package input with a default	It appears on new proposals without a new DocType field or code change.
Conditional panel-count bands / distance thresholds	Use only the configured matching branch, including exact boundary values.
Compatible hybrid: battery on / off	Battery and Battery DB templates are included / excluded together.
Package does not allow battery	include_battery is false; no battery templates are generated.
Missing input, misspelled key, cycle or invalid quantity	Show the affected template and row; do not partially replace a saved proposal.
No valuation or missing UOM conversion	Material calculation remains inspectable; cost status is unresolved rather than silently priced at zero.

Solar Proposal: inputs, generated rows and saved valuation estimates

Choose the five main inputs, complete package-defined site values, then generate and review the proposal. Battery is optional on compatible hybrids.

Solar Proposal — staff-entered and defaulted fields

Field / internal field name	Frappe type / linked DocType	Who fills it?	Example / purpose
Party Type / Party proposal_to / party	Link → DocType / Dynamic Link	Staff selects the existing Customer, Lead or Prospect	Existing party flow is retained.
Proposal Date proposal_date	Date	Defaults today; staff may change	The proposal document date.
System / Package package_name	Link → Solar Package	Main input 1	A configured system profile.
Number of Phases connection_type	Select	Main input 2	Single Phase / Three Phase; available as phase in expressions.
Down Run Cabling down_run_cabling	Select	Main input 3	Armoured Cabling / Conduit Cabling; available as cabling_type.
System Capacity capacity_kw	Float	Main input 4	Requested system capacity in kW.
Panel Capacity panel_capacity_w	Int	Main input 5	Selected panel wattage.
Include Battery include_battery	Check	Only shown / selectable when the package allows it	Enables the battery-related template mappings.
Additional Inputs input_values	Table → Solar Proposal Input Value	Defaults from package definitions; staff edits site values	Lengths, screw choice, battery count and other configured variables.
Panel Brand / Include Structure panel_brand / include_structure	Select / Check	Existing staff choices	Retain current fields; see the costing notes on tab 6.
Company company	Link → Company	Defaulted from settings / ERPNext	Company context and base currency for valuation.
Valuation Warehouse valuation_warehouse	Link → Warehouse	Defaulted from settings; staff confirms	A non-group warehouse belonging to the selected company.
Discount / Subsidy discount / subsidy	Currency / Currency	Existing inputs / subsidy default behavior	No new discount, subsidy or margin model in this stage.
Remarks remarks_field	Small Text	Optional staff entry	Site-specific information.

Solar Proposal — generated / saved result fields

Field / internal field name	Frappe type / linked DocType	Who fills it?	Example / purpose
Customer Name customer_name	Data	Auto from party	Existing behavior.
Panel Count panel_count	Int	Auto: ceil(capacity_kw × 1000 / panel_capacity_w)	3 kW / 660 W → 5 panels.
Inverter Count inverter_count	Int	Auto from the package count formula	The count variable used by this package’s material rules.
Installed Panel Capacity actual_capacity_kw	Float	Auto: panel_count × panel_capacity_w / 1000	5 × 660 W → 3.3 kWp.
Material List table_proposal_bom	Table → Solar Proposal BOM Item	Auto after conditions, dependencies and quantities succeed	Actual Items from all applicable package templates.
Existing Other Charges cost_breakdown	Table → Project Cost Breakdown	Copy existing defaults and calculate current charges	Staff can review and edit current charge rows.
Material / Other Cost material_cost / other_cost	Currency / Currency	Auto	Valued material amounts / included existing charges.
Project / Final Amount project_cost / final_amount	Currency / Currency	Auto	Materials + other charges; then existing discount / subsidy deductions.
Currency / Cost Status currency / costing_status	Link → Currency / Select	Auto	Company currency; Complete or Missing Valuation / Conversion.
Calculation Definition Snapshot configuration_snapshot	JSON — system-managed	Auto when generated; retained on ordinary save / reopen	Saved package formula, input definitions, template mapping conditions and material rules used to generate this proposal.

Save the input values, chosen Item mappings, quantities, rule explanations and rates with the proposal. Editing a template, an Item or a package default does not silently rewrite those saved results.

Solar Proposal BOM Item — existing material child, extended for traceability

Field / internal field name	Frappe type / linked DocType	Who fills it?	Example / purpose
Group / Generic Label component_group / component_label	Data / Data	Copied from the template and rule	The role/group used to explain and print the material.
Source Template / Alias / Rule source_template / template_alias / rule_key	Link → Solar Component Template / Data / Data	Auto	A stable trace back to the definition that generated the row.
Item item_code	Link → Item	Copied automatically from the template	The actual Item staff selected once during template setup.
Item Name / Specification item_name / description	Data / Small Text	Saved copies	The material text used by the proposal PDF.
Quantity / UOM quantity / uom	Float / Link → UOM	Calculated from the rule; staff may make a recorded site override	Quantity expressed in the template row UOM.
Generated Quantity / Override Flag generated_quantity / has_quantity_override	Float / Check	Auto; preserve the calculated basis and record an override	If a clamp quantity is overridden, dependent screw / plug formulas use the effective overridden quantity.
Valuation Rate rate	Currency	Auto; per row UOM	Stock-unit valuation converted to the material UOM.
Amount amount	Currency	Auto	quantity × rate.
Calculation Note calculation_note	Small Text	Auto snapshot	Formula and substituted values explaining the result.
Rate Status / Override Reason valuation_status / override_reason	Select / Small Text	Auto status; staff explains any site override	Missing valuation is visibly different from a confirmed zero-rate row.

Site quantity overrides apply to included rows, are recorded with a reason, and feed dependent calculations. For example, overriding clamps from 50 to 60 makes a two-screws-per-clamp formula produce 120 screws. Excluded rows stay at zero. Ordinary recalculation uses saved definitions and rates; it does not reload edited master records.

Regenerate deliberately reloads current definitions and replaces the generated snapshot only after the complete calculation succeeds. It preserves site-entered input values, adds defaults for new inputs, and explicitly resets direct material-quantity overrides. Reprice refreshes valuation estimates without changing material quantities. Ordinary save / reopen preserves the existing snapshots and overrides.

All evaluations and final totals use one server calculation service. The form previews the same service; API-created proposals and browser edits must pass the same validation.

Keep existing charges; use valuation estimates for materials

This stage builds the configurable material engine. Price lists, new markup methods and margin redesign are outside this stage.

Wahni Default Settings — existing charge defaults and valuation context

Field / internal field name	Frappe type / linked DocType	Who fills it?	Example / purpose
Cost Calculation table_cost_calculation	Table → Project Cost Breakdown	Staff maintains existing default rates	The same ten cost-component options remain.
Subsidy Rules table_subsidy_rule	Table → Subsidy Rule	Staff maintains existing ranges / amounts	Keep the current subsidy behavior.
Existing Panel Settings panel_min_capacity / panel_max_cacpacity	Int / Int	Existing values retained	The current Panel Cost calculation uses panel_max_cacpacity.
Default Costing Company costing_company	Link → Company	Staff sets a default	One company/base-currency context for material valuation.
Default Valuation Warehouse valuation_warehouse	Link → Warehouse	Staff sets a non-group warehouse in that company	No arbitrary warehouse selection or hidden buying-price fallback.

Project Cost Breakdown — retain the current six fields

Field / internal field name	Frappe type / linked DocType	Who fills it?	Example / purpose
Cost Component cost_component	Select	Defaults from settings; staff may change	One of the existing ten components below.
Rate rate	Currency	Staff maintains defaults / proposal adjustments	Rate basis depends on the existing component.
Quantity quantity	Float	Current automatic default; staff may edit	Panel count, ceil(panel count / 2), requested kW or 1.
Included included	Check	Existing default; staff may toggle	Unchecked → zero amount.
Amount amount	Float — existing field type	Auto	Current formula for the chosen cost component.
Calculation Note calculation_note	Data	Auto explanation	Show the units / basis used by the charge.

Current charge formulas — no margin redesign

Cost Component	Default quantity	Rate basis	Current included amount
Panel Cost	panel_count	Per watt	rate × settings.panel_max_cacpacity × panel_count
Structure Material	ceil(panel_count / 2)	Per two-panel set	rate × quantity
Labour Structural	ceil(panel_count / 2)	Per two-panel set	rate × quantity
Labour Electrical	ceil(panel_count / 2)	Per two-panel set	rate × quantity
Supervisor Documentation and Visits	1	Per project	rate
Sales Commission	capacity_kw	Per requested kW	rate × capacity_kw
Shipping	1	Per project	rate
Margin	1	Fixed amount per proposal	rate; no percentage / markup model added
Walk Way	0; optional staff entry	Staff’s chosen quote basis	rate × quantity when included
Ladder	0; optional staff entry	Staff’s chosen quote basis	rate × quantity when included

Current-code notes: Panel Cost uses the global maximum panel wattage, and Include Structure does not currently switch off structure charge rows. The previous table draft proposed changing those behaviors. They are documented here as costing follow-ups, not silently presented as already implemented or added to this material-engine stage.

Material valuation and customer PDF steps

Step	Source / action	Calculation / check	Result
1 — Generate materials	Selected package, inputs, templates and actual Item links	Evaluate conditions and quantity dependencies	A complete material snapshot; no partial replacement on formula errors.
2 — Value materials	Company / warehouse stock valuation	Validate stock-to-row UOM conversion; amount = quantity × row-unit rate	Saved valuation estimate; unresolved rates / conversions are flagged.
3 — Add current charges	Existing Project Cost Breakdown	project_cost = material_cost + other_cost	Keep existing charge defaults and formulas.
4 — Apply current deductions	Existing discount and subsidy fields	final_amount = project_cost − discount − subsidy	The existing proposal price summary.
5 — Review and print	Save → Print → Solar Proposal → PDF	Resolve formula and costing errors before final output	Customer system details, grouped material list, price summary and existing terms.

Use valuation only: no silent standard-rate or buying-price fallback. Convert stock UOM rates into the material UOM; for arithmetic illustration, ₹100 per metre × 25 metres per roll gives ₹2,500 per roll. Missing conversion is an unresolved cost, not a conversion factor guessed as 1.

Empty templates and omitted workbook components remain editable. Formula errors prevent generation; missing valuation may remain in an incomplete draft but must be resolved before the customer PDF. Internal costs / margin rows do not need to appear in the customer PDF; existing Solar Proposal Content supplies warranty and terms.